

Enhancing Discrete Choice Models with Representation Learning

A Hybrid Approach Combining DCM and Neural Networks

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Course: Discrete Choice Modeling
May 2026

Based on Sifringer, Lurkin & Alahi (2020) - Transportation Research Part B: Methodological

Presentation Roadmap

- 1 Introduction & Motivation
- 2 Background & Preliminaries
- 3 Proposed Methodology
- 4 Experimental Setup
- 5 Synthetic Data Experiments
- 6 Real Case Study: Swissmetro
- 7 Small Dataset Case: Optima
- 8 Key Theoretical Insights
- 9 Comparison with Prior Work
- 10 Practical Guidelines
- 11 Future Directions
- 12 Conclusions

The Core Tension in Choice Modeling

Two Competing Paradigms

- **Discrete Choice Models (DCMs)**: High interpretability, but potential misspecification.
- **Neural Networks (NNs)**: High predictive power, but often seen as a "black box".

Research Question

Can we combine the **interpretability** of DCM with the **predictive power** of representation learning?

Key Insight: Divide utility into a knowledge-driven (interpretable) part and a data-driven (learned) part.

Why is this Important?

- **Misspecification Problem:** Standard DCMs assume utility is known a priori (e.g., linear-in-parameters).
- **Consequences:**
 - Biased parameter estimates.
 - Poor out-of-sample prediction.
 - Incorrect willingness-to-pay / Value of Time (VOT).
- **Limitation of Pure ML:** Neural networks lack interpretability (no elasticities, no t-tests, no VOT).

Goal: L-MNL and L-NL models to bridge this gap.

Quick Recap: Multinomial Logit (MNL)

Random Utility Maximization (RUM)

$$U_{in} = V_{in} + \varepsilon_{in}$$

V_{in} : Systematic utility, ε_{in} : i.i.d. Extreme Value Type I.

Linear-in-Parameters Utility

$$V_{in} = \sum_d \beta_d \cdot x_{din}$$

Choice Probability

$$P_n(i) = \frac{e^{V_{in}}}{\sum_{j \in C_n} e^{V_{jn}}}$$

MNL as a Neural Network

- A neural network maps input space x to output U through hidden layers $h^{(j)}$:

$$\mathbf{U} = h^{(L)}(q^{(L-1)}), \quad q^{(j)} = h^{(j)}(q^{(j-1)}), \quad \forall j = 1, \dots, L, \quad q^{(0)} = x$$

- Using a CNN, the neuron i in layer $(j+1)$:

$$h_i^{(j+1)} = g\left(\sum_{k=0}^d h_{(s \cdot i + k)}^{(j)} \beta_k^{(j)} + \alpha_i^{(j)}\right)$$

where $\{\beta_1, \dots, \beta_d\} = \beta$ is the filter, s is the stride, α_i a bias, and $g(\cdot)$ the activation function.

- MNL probabilities are obtained with softmax:

$$(\sigma(\mathbf{V}_n))_i = \frac{e^{V_{in}}}{\sum_{j \in C_n} e^{V_{jn}}}$$

- Loss: Categorical cross-entropy

$$H_n(\sigma, y_n) = - \sum_{i \in C_n} y_{in} \log[\sigma_i(\mathbf{V}_n)]$$

MNL as a Neural Network

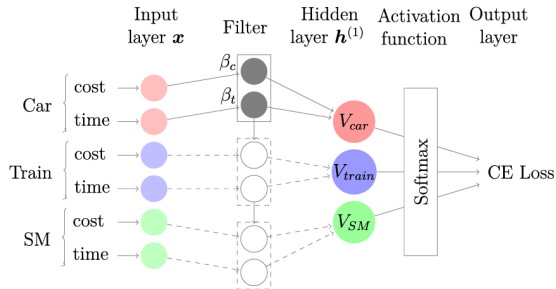


Figure: MNL implemented as a 1-layer CNN with identity activation and softmax output.

- Input: Alternative-specific variables (Cost, Time).
- Filter: β parameters.
- Output: Softmax gives MNL probabilities.
- Loss: Categorical Cross-Entropy (equivalent to Log-Likelihood).

General Framework: Hybrid Utility

The Core Equation

$$V_{in} = \underbrace{f_i(\mathbf{X}_n; \beta)}_{\text{Knowledge-Driven (Interpretable)}} + \underbrace{r_i(\mathbf{Q}_n; \mathbf{w})}_{\text{Data-Driven (Representation Learning)}}$$

- $\mathbf{X}$: Variables with known relationships (e.g., cost, time).
- $\mathbf{Q}$: Variables with unknown/complex relationships (e.g., images, text, or raw socio-demographics).
- β : Interpretable parameters (t-stats available).
- $\mathbf{w}$: Neural network weights (not interpreted directly).

Interpretability Condition

To preserve DCM interpretability for a variable t in $\mathbf{X}$:

$$\frac{\partial r_{in}}{\partial t_{in}} = 0$$

- The neural network cannot learn a direct effect of variables that are supposed to be interpreted.
- If a variable is in both $\mathbf{X}$ and $\mathbf{Q}$, the elasticity becomes:

$$\frac{\partial U_{in}}{\partial t_{in}} = \frac{\partial f_{in}}{\partial t_{in}} + \frac{\partial r_{in}}{\partial t_{in}}$$

Conclusion: $\mathbf{X}$ and $\mathbf{Q}$ must be disjoint (or low correlation).

Error decomposition:

$$\bar{\varepsilon}_{in} = r_i(\mathbf{Q}_n; \mathbf{w}) + \varepsilon_{in}$$

This separates the new representation term $r_i(\mathbf{Q}_n; \mathbf{w})$ from the original error ε_{in} .

L-MNL: Learning Multinomial Logit

- $r_i(\mathbf{Q}_n; \mathbf{w})$ is a Dense Neural Network (ReLU activations).
- The final choice probability:

$$P_n(i) = \frac{e^{f_i(\mathbf{X}_n; \beta) + r_i(\mathbf{Q}_n; \mathbf{w})}}{\sum_j e^{f_j(\mathbf{X}_n; \beta) + r_j(\mathbf{Q}_n; \mathbf{w})}}$$

- The representation term r_i as a DNN with L layers of H neurons:

$$r_{in} = \sum_{k=1}^H w_{ik}^{(L)} g(\mathbf{q}_n^{(L-1)} \mathbf{w}_k^{(L-1)} + \alpha_k^{(L-1)}) + \alpha_i^{(L)}$$

- Recurrent definition of intermediate layers:

$$[\mathbf{q}_n^{(j+1)}]_i = \sum_{k=1}^H w_{ik}^{(j+1)} g(\mathbf{q}_n^{(j)} \mathbf{w}_k^{(j)} + \alpha_k^{(j)}) + \alpha_i^{(j+1)}$$

- Input layer: $\mathbf{q}_n^{(0)}$ is the vector of input features $\mathbf{Q}_n$.

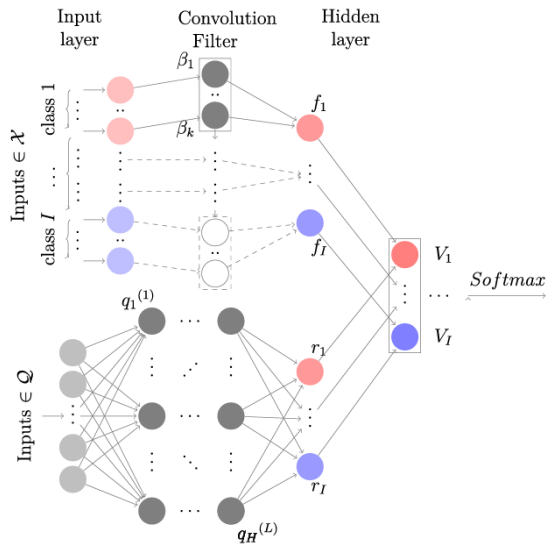


Figure: L-MNL Architecture

Extension: L-NL (Learning Nested Logit)

- Relaxes the IIA assumption by grouping correlated alternatives into nests.
- The probability for alternative i in nest m :

$$P_n(i|C) = \underbrace{\frac{e^{\mu_m V_{in}}}{\sum_{j \in C_m} e^{\mu_m V_{jn}}}}_{\text{Within nest}} \times \underbrace{\frac{\exp\left(\frac{\mu}{\mu_m} \ln \sum_{p \in C_m} e^{\mu_m V_{pn}}\right)}{\sum_k \exp\left(\frac{\mu}{\mu_k} \ln \sum_{p \in C_k} e^{\mu_k V_{pn}}\right)}}_{\text{Between nests}}$$

- **L-NL:** Replace V_{in} with $f_i + r_i$ and estimate nest parameters μ_m via custom NN layers.

Benchmarking Models Compared

Model	Specification	Parameters
$\text{Logit}(\chi)$	Standard MNL, no learning	$ \chi $
$\text{DNN}(n, Q)$	Pure NN (Hruschka et al.)	$n(Q + C + 1)$
$\text{DNN_L}(n, \chi=Q)$	Residual NN (ResNet style)	$(n + 1)(\chi + C)$
L-MNL (n, χ, Q)	Proposed (disjoint sets)	$ \chi + n(Q + C)$

Table: Models compared in the paper

- All NN models: 20% dropout, Adam optimizer, 200 epochs.
- Synthetic, Semi-synthetic, and Real datasets.

Synthetic Data Generation

True utility (with unknown interaction):

$$V_{in} = \beta_p p_{in} + \beta_a a_{in} + \beta_b b_{in} + \underbrace{\beta_{qc} q_{in} c_{in}}_{\text{Hidden non-linearity}}$$

- $\mathcal{X} = \{p, a, b\}$ (observed by modeler).
- $\mathcal{Q} = \{q, c\}$ (given to NN).
- True parameters: $\beta_p = -1, \beta_a = 0.5, \beta_b = 0.5, \beta_{qc} = 1$.
- Goal: Recover β_p, β_a accurately while predicting well.

Monte Carlo Results: Log-Likelihood

Model	Train LL	Test LL	Accuracy %
Logit(χ_{true})	-459	-94	87
Logit(χ_1)	-604	-123	66
DNN(25, Q)	-363	-112	84
DNN_L(25, $\chi=Q$)	-367	-108	83
L-MNL(25, χ, Q)	-429	-97	87

Table: Monte Carlo average log-likelihood (100 experiments)

Observation: L-MNL achieves near-oracle test performance despite not seeing the interaction term directly.

Parameter Recovery Results

Relative Error Formulas

$$e_{\beta} = \frac{|\hat{\beta} - \beta|}{\beta}, \quad e_{\beta_i/\beta_j} = \frac{e_{\beta_i} - e_{\beta_j}}{1 - e_{\beta_j}}$$

Model	ϵ_{β_p} (%)	ϵ_{β_a} (%)	$\epsilon_{\beta_p/\beta_a}$ (%)
Logit(χ_{true})	6.4	14.4	10.8
Logit(χ_1)	26.7	26.7	15.5
DNN_L(25, $\chi=Q$)	60.0	74.0	460.0
L-MNL(25, χ, Q)	7.1	15.2	11.3

Table: Relative errors in parameter estimates

Important: DNN_L (ResNet style) completely fails to recover β because $\partial r / \partial x \neq 0$.

Hypothesis Testing Results

Model	$H_0 : \beta_p = -1$	$H_0 : \beta_a = 0.5$	$H_0 : \beta_p/\beta_a = -2$
Logit(χ_{true})	97.5%	97.5%	94%
Logit(χ_1)	34%	34%	97%
DNN_L	25.5%	25.5%	37%
L-MNL	95%	95%	94%

Table: Percentage of experiments not rejecting null hypothesis (95% CI)

- L-MNL correctly identifies true parameters as often as the true MNL.
- Logit(χ_1) fails on individual coefficients but recovers ratios (scale invariance of logit).

Effect of NN Architecture Size

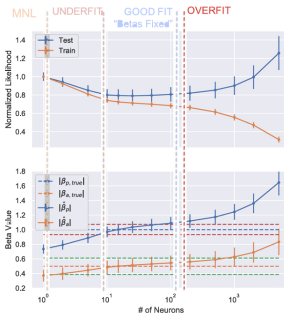


Figure: Parameter estimates vs. NN size

- $n = 0$ (MNL): Biased due to misspecification.
- $n = 10$ to 100 : True parameters recovered.
- $n > 500$: Overfitting degrades interpretable parameters.

Impact of Correlated Variables (Critical Warning)

- Condition: $\frac{\partial r}{\partial x} = 0$ requires $\mathbf{X}$ and $\mathbf{Q}$ to be statistically independent.
- Experiment: Correlate $p \in \mathcal{X}$ with $q' \in \mathcal{Q}$.
- Correlation coefficient s :

$$q'_{in} = s \cdot p_{in} + \sqrt{1 - s^2} \cdot q_{in}$$

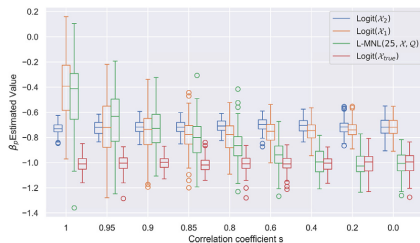


Figure: Impact of correlated variables on parameter estimates

Practical Advice: Avoid including correlated variables across $\mathcal{X}$ and $\mathcal{Q}$.

Swissmetro Dataset

- Stated Preference survey, Switzerland (1998).
- Modes: Car, Train, Swissmetro (new maglev).
- $N = 9,036$ observations after cleaning.
- Variables:

$\mathcal{X}$ (Interpretable)	$\mathcal{Q}$ (NN Input)	Description
TT (Travel Time)	Purpose, First, Ticket	Trip context
Cost	Who, Male, Income	Demographics
Freq (Frequency)	Origin, Dest, GA, Luggage, Age, Seats	Other attributes

Swissmetro: Model Comparison

Model	β_{cost} (t-stat)	β_{time} (t-stat)	Test LL	ρ^2
MNL (Bierlaire)	-0.695 (-16.4)	-	-1433	0.28
L-MNL(χ_1 , $\mathcal{Q}_1$)	-1.378 (-28.4)	-1.563 (-27.9)	-1181	0.41
DNN.L(χ_1 , $\mathcal{Q}_1$)	-0.897 (-19.4)	-1.146 (-23.3)	-1257	0.37
Logit(all 41 inputs)	-1.062 (18.0)	-1.326 (25.0)	-1322	0.33
L-MNL(χ_2, $\mathcal{Q}_2$)	-1.671 (-31.9)	-1.769 (-45.4)	-1108	0.44

Table: Parameter estimates and performance on Swissmetro

Note: $\chi_2 = \{\text{Time, Cost, Freq}\}$, $\mathcal{Q}_2 = \text{all other variables}$.

Swissmetro: Value of Time (VOT) Improvements

Model	VOT (CHF/min)	Test LL
Logit(λ_1)	0.52	-1433
Logit(λ_2)	0.80	-1322
DNN_L	0.78	-1257
L-MNL(λ_1)	0.88	-1181
L-MNL(λ_2)	0.94	-1108
CNL (Cross-Nested)	0.59	-1415
TPM (Panel Mixture)	0.72	-1350

Table: VOT comparison across models

Observation: As model flexibility increases, VOT moves toward 0.94 CHF/min (likely the true behavioral value).

L-NL Results: Nesting Correlation

Model	Nest Parameter μ	Test LL
NL (λ_1 , no Q)	0.60	-1415
L-NL (λ_1 , Q_0)	0.75	-1380
L-NL (λ_1 , Q_1)	0.90	-1300
L-NL (λ_2 , Q_2)	1.00	-1200

Table: Nest parameter trends

- Nest parameter $\mu = 1$ implies no remaining correlation between Car & Train.
- Interpretation: The representation term absorbed the misspecification that originally caused IIA violation.

Swissmetro: Sensitivity Analysis (Interpretability)

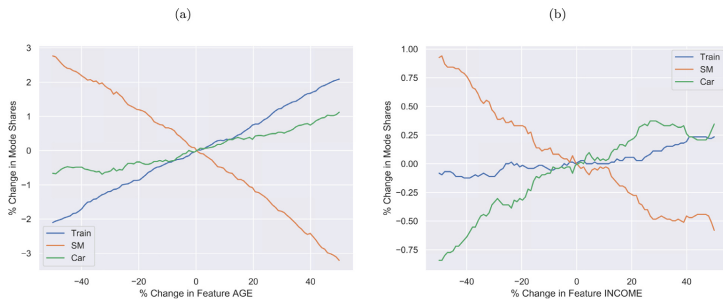


Figure: Percentage change in total mode shares against percentage change in L-MNL features

- Even though Age is in $\mathcal{Q}$, we can still analyze its impact via saliency maps or sensitivity analysis.
- Age shows near-linear effect, Income shows non-linear effect.

Optima Dataset: Small Revealed Preference

- Survey in Switzerland (2009-2010).
- $N = 1,376$ (very small for deep learning).
- Alternatives: Public Transport, Car, Slow modes.
- Benchmark: ICLV (Integrated Choice and Latent Variable) model from Fernández-Antolín et al. (2016).

Model	Train Accuracy	Test Accuracy
MNL(χ_1)	76.8%	76.7%
L-MNL(χ_1, Q_1)	80.4%	79.2%
NN (40 neurons)	86.1%	81.3%
ICLV (Latent Variable)	80.0%	77.7%

Table: Optima dataset results

Expert Modeling as a Regularizer

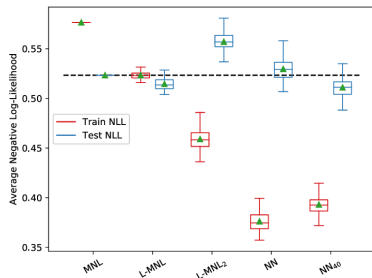


Figure: Performance of multiple models on a small revealed preference dataset

- Pure NN (40 neurons) achieves highest test accuracy (81.3%) but with high variance and no interpretability.
- L-MNL provides good accuracy (79.2%) with full interpretability (t-stats, elasticities).
- **Key takeaway for small data:** The DCM part acts as a strong regularizer.

Correcting Misspecification Bias

Without representation term

$$\bar{\varepsilon}_{in} = r_i(\mathbf{Q}_n; \mathbf{w}) + \varepsilon_{in}$$

The error term contains systematic structure $\rightarrow$ biased β .

With representation term

$$U_{in} = f_i(\mathbf{X}_n; \beta) + r_i(\mathbf{Q}_n; \mathbf{w}) + \varepsilon_{in}$$

- NN absorbs systematic patterns from $\mathbf{Q}$.
- Remaining ε_{in} is closer to i.i.d. $\text{EV}(0,1)$.
- β becomes unbiased (if $\mathbf{X} \perp \mathbf{Q}$).

Why Joint Optimization Matters

Strategy	$\hat{\beta}_p$	$\hat{\beta}_a$	Test LL
(1) β then w	-1.41	0.75	-942
(2) w then β	-1.59	0.82	-758
(3) Joint (proposed)	-1.95	1.00	-721

Table: Optimization strategies for L-MNL (true $\beta_p = -2, \beta_a = 1$)

Conclusion: Sequential estimation fails. The two components must be trained simultaneously via backpropagation.

Relation to Existing Hybrid Models

Model	Formulation	Our Framework
Bentz & Merunka (2000)	$V = 0 + r(Q)$	$\mathcal{X} = \emptyset$
Hruschka (2007)	$V = f(X) + r(X)$	$\mathcal{X} = \mathcal{Q}$ (violates interpretability)
Wang et al. (2020)	Alternative-specific NN	Special case of our general framework
Ours (L-MNL)	$V = f(\mathcal{X}) + r(\mathcal{Q})$	$\mathcal{X} \cap \mathcal{Q} = \emptyset$

Table: Positioning L-MNL in the literature

How to Use L-MNL in Practice

- 1 **Define $\mathcal{X}$:** Variables for which you need interpretable β (e.g., Cost, Time to compute VOT).
- 2 **Define $\mathcal{Q}$:** All other variables (income, age, trip purpose, images, text, etc.).
- 3 **Check correlation:** Ensure $\mathcal{X}$ and $\mathcal{Q}$ are not highly correlated (use VIF or correlation matrix).
- 4 **Choose NN size:** Start small (10-50 neurons), monitor test LL, stop before overfitting.
- 5 **Jointly optimize:** Use Adam optimizer, categorical cross-entropy loss.
- 6 **Interpret β :** Standard errors from Hessian approximation (same as MNL).

When Does L-MNL Fail?

- **High correlation** between $\mathcal{X}$ and $\mathcal{Q}$ (biases β).
- **Very small datasets** ($N < 500$): Pure NN overfits, but L-MNL still helps as regularizer.
- **Endogeneity in $\mathcal{X}$** : L-MNL does not fix endogeneity (use control functions or IV).
- **Complex substitution patterns**: Use L-NL instead of L-MNL.

Future Research Avenues

- **Other DCM kernels:** Mixed Logit, Latent Class, GEV models with representation learning.
- **Non-tabular $\mathcal{Q}$:** Images (CNNs), sequences (RNNs/Transformers), graphs (GNNs).
- **Automatic feature discovery:** Use NN to propose new interaction terms for $\mathcal{X}$.
- **Explainable AI integration:** Saliency maps, SHAP, LIME to interpret $r(\mathcal{Q})$.
- **Bidirectional benefits:** DCM as a regularizer for ML on small datasets.

Main Takeaways

- ❶ **Problem:** DCMs suffer from misspecification; pure NNs lack interpretability.
- ❷ **Solution:** L-MNL / L-NL = Interpretable linear part + Learned residual part.
- ❸ **Condition:** $\frac{\partial r}{\partial x} = 0 \Rightarrow \mathcal{X}$ and $\mathcal{Q}$ must be disjoint/low correlation.
- ❹ **Results:**
 - Synthetic: Recovers true β as well as oracle MNL.
 - Swissmetro: Test LL improves from -1433 to -1108, VOT becomes plausible.
 - Optima: Outperforms ICLV on small data with interpretability.
- ❺ **Code:** Open source available (promotes reproducible research).

Take-Home Message for Practitioners

"Do not choose between interpretability and predictive power.
Use both."

- Start with a good DCM specification ($\mathcal{X}$).
- Add a neural network to learn the rest ($\mathcal{Q}$).
- Jointly optimize.
- Get better predictions AND interpretable elasticities.

References

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- ❹ Hruschka, H., Fettes, W., & Probst, M. (2004). An empirical comparison of neural net based multinomial logit. *European Journal of Operational Research*, 159(1), 166-180.

Code: <https://github.com/vita-epfl/learning-choice-models>

Thank You

Questions?

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May 2026
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